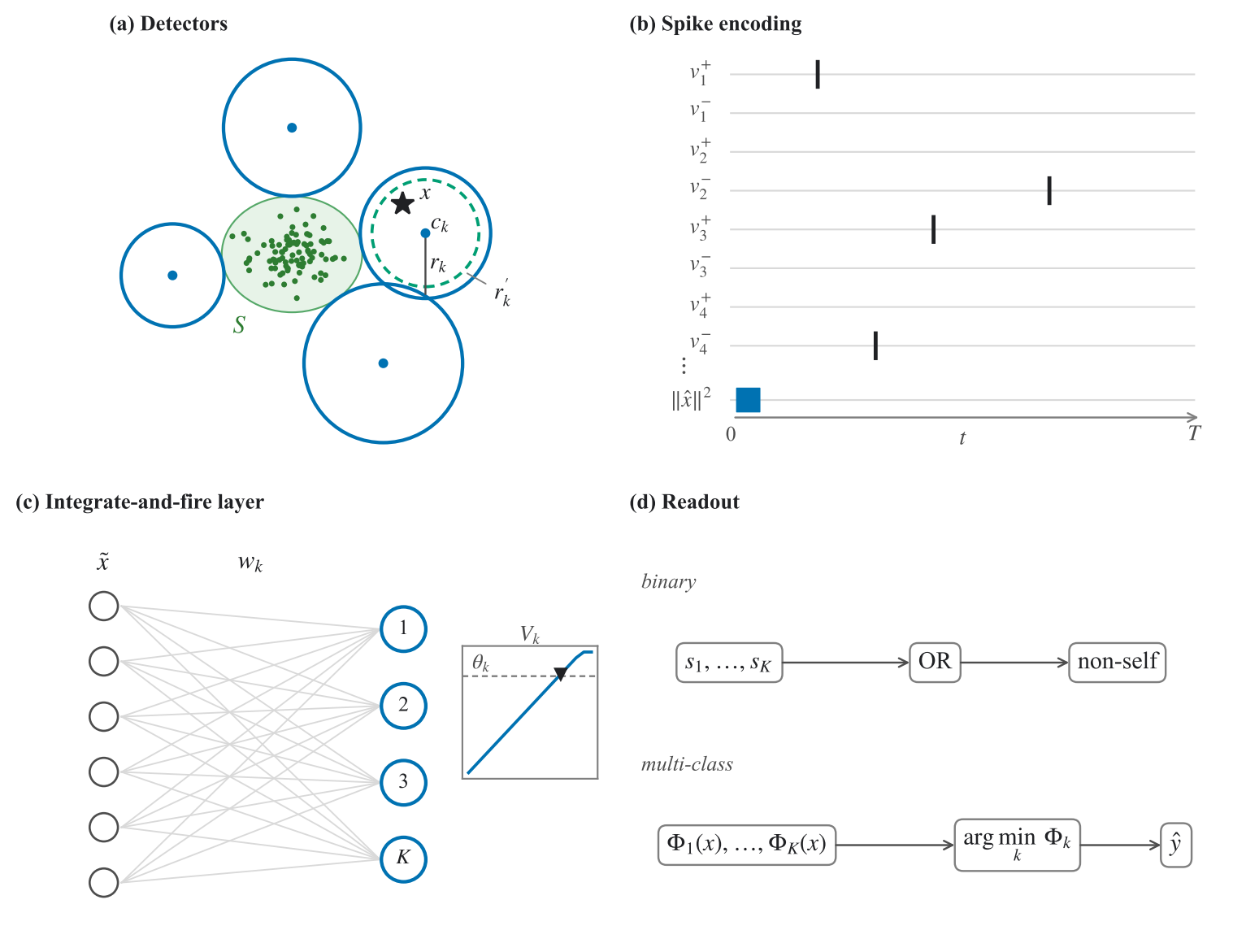
(a) Detectors
x
c_k
r_k
r_k'
S
(b) Spike encoding
v_1^+
v_1^-
v_2^+
v_2^-
v_3^+
v_3^-
v_4^+
v_4^-
$\|\hat{x}\|^2$
0
t
T
(c) Integrate-and-fire layer
$\tilde{x}$
w_k
1
2
3
K
V_k
θ_k
(d) Readout
binary
$s_1, \ldots, s_K$
OR
non-self
multi-class
$\Phi_1(x), \ldots, \Phi_K(x)$
$\arg\min_k \Phi_k$
$\hat{y}$